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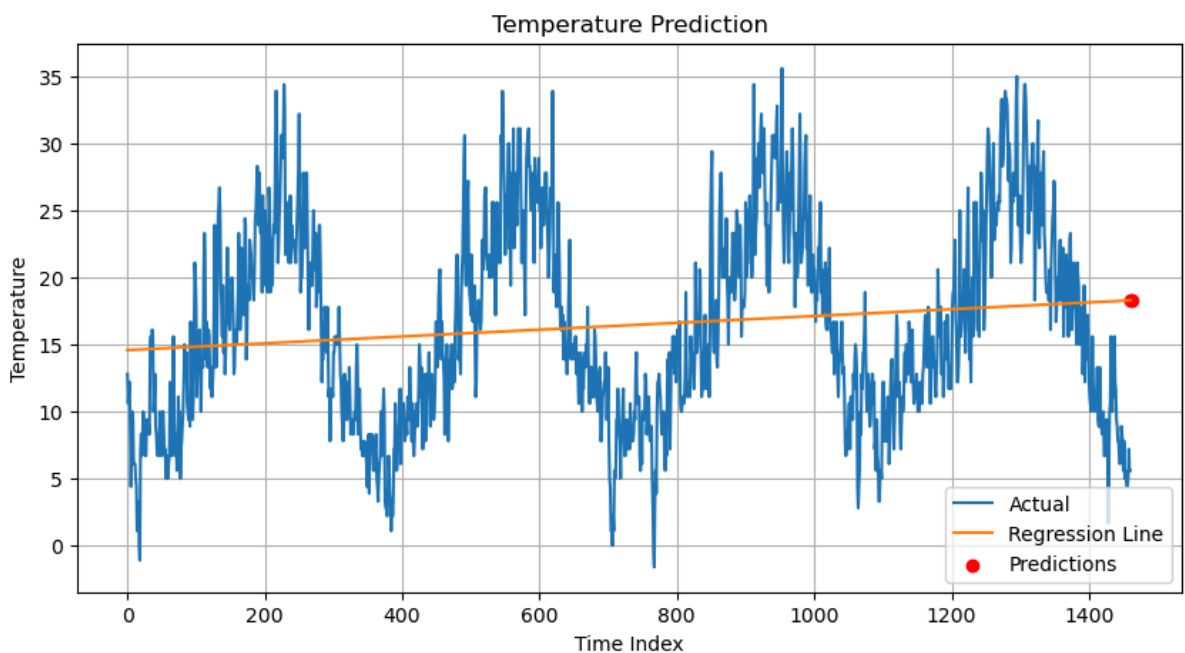
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
url = r"C:\Users\CAD LAB 04\Desktop\334\seattle-weather.csv"
data = pd.read_csv(url)
print("Columns:", data.columns)
data = data.dropna()
temp_col = None
for col in data.columns:
    if "temp" in col.lower():
        temp_col = col
        break
if temp_col is None:
    raise Exception("No temperature column found!")
print("Using temperature column:", temp_col)
data['TimeIndex'] = range(len(data))
X = data[['TimeIndex']]
y = data[temp_col]
model = LinearRegression()
model.fit(X, y)
future = pd.DataFrame({
    'TimeIndex': [len(data), len(data) + 1]
})
pred = model.predict(future)
print("Predicted:", pred)
plt.figure(figsize=(10, 5))
plt.plot(data['TimeIndex'], y, label="Actual")
plt.plot(data['TimeIndex'], model.predict(X), label="Regression Line")
plt.scatter(future['TimeIndex'], pred, color='red', label="Predictions")
plt.xlabel("Time Index")
plt.ylabel("Temperature")
plt.title("Temperature Prediction")
plt.legend()
plt.grid(True)
plt.show()

```

Columns: Index(['date', 'precipitation', 'temp_max', 'temp_min', 'wind', 'weather'], dtype='object')

Using temperature column: temp_max

Predicted: [18.29612285 18.29866326]



In []: SE-A,S3
ROLL NO-341